

8.2 Diffraction & Interference

Question Paper

Course	CIE A Level Physics
Section	8. Superposition
Topic	8.2 Diffraction & Interference
Difficulty	Medium

Time allowed: 60

Score: /44

Percentage: /100

Question 1a

Fig. 1.1 shows an arrangement for observing the interference pattern produced by laser light passing through two narrow slits S_1 and S_2 .

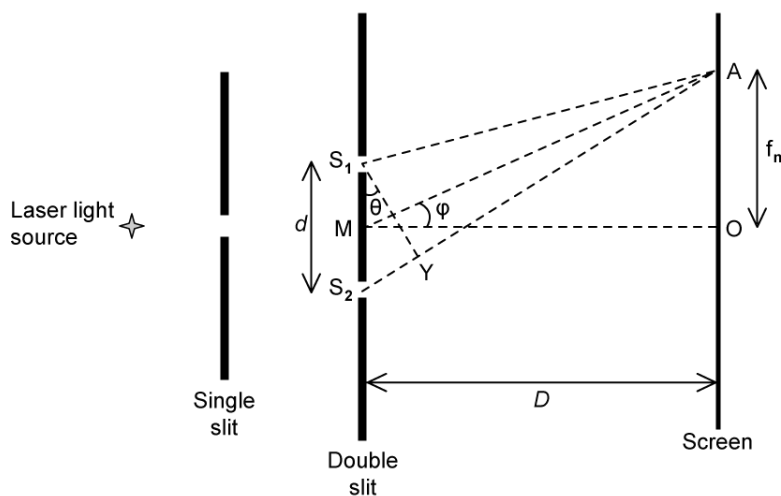


Fig. 1.1

The distance S_1S_2 is d , and the distance between the double slit and the screen is D , where $D \gg d$, so angles θ and ϕ are small.

M is the midpoint of S_1S_2 and it is observed that there is a bright fringe at point A on the screen, a distance f_n from point O on the screen. Light from S_1 travels a distance S_2Y further to point A than light from S_1 .

The wavelength of light from the laser is 650 nm and the angular separation of the bright fringes on the screen is $5.00 \times 10^{-4} \text{ rad}$.

Calculate the distance between the two slits.

[3 marks]

Question 1b

A bright fringe is observed at A.

(i)
Explain the conditions required in the paths of the rays coming from S_1 and S_2 to obtain this bright fringe.

[2]

(ii)
State an equation in terms of wavelength for the distance S_2Y .

[1]

[3 marks]**Question 1c**

Deduce expressions for the following angles in the double-slit arrangement shown in part a:

(i)
 θ in terms of S_2Y and d

[2]

(ii)
 ϕ in terms of D and f_n

[2]

[4 marks]

Question 1d

The separation of the slits S_1 and S_2 is 1.30 mm.

The distance MO is 1.40 m.

The distance f_n is the distance of the ninth bright fringe from O and the angle θ is 3.70×10^{-3} radians.

Calculate the wavelength of the laser light.

[2 marks]

Question 2a

A student is conducting a series of investigations on diffraction and interference with two slits.

In the first investigation, monochromatic light passes through a double-slit arrangement. The intensity of the fringes varies with distance from the central fringe. This is observed on a screen, as shown in Fig. 1.1.

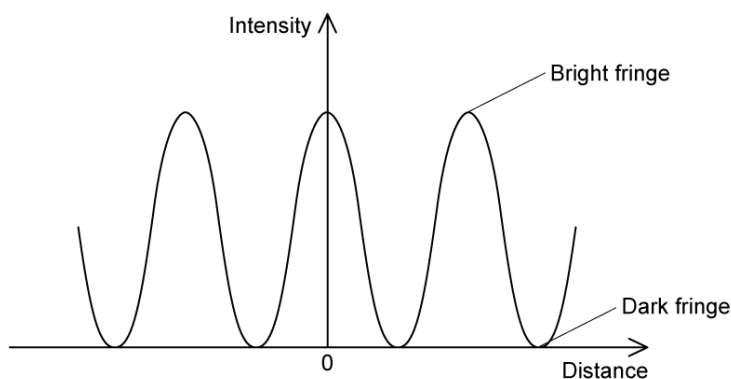


Fig. 1.1

The intensity of the monochromatic light passing through one of the slits is reduced.

Explain the effect of this change on the appearance of

(i)
The dark fringes.

[3]

(ii)
The bright fringes.

[3]

[6 marks]

Question 2b

In an investigation, a white light source is incident on an orange filter, a single slit, and then a double-slit, as shown in Fig. 1.2.

An interference pattern of light and dark fringes is observed on the screen.

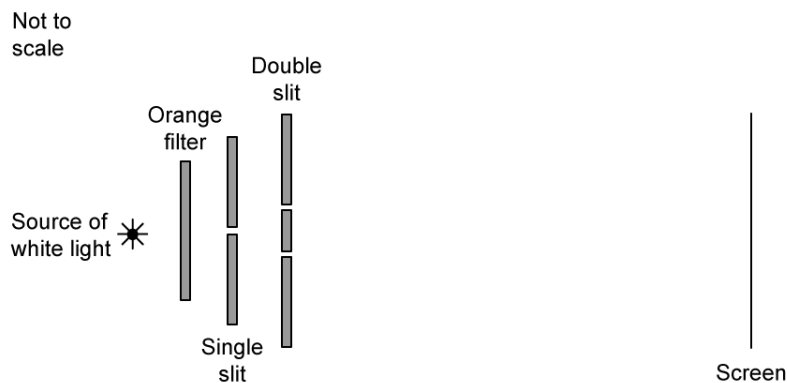


Fig. 1.2

(i)

The orange filter is now replaced by a green filter.

State and explain the change in appearance, other than the change in colour, of the fringes on the screen.

[2]

(ii)

The green filter is now removed.

State and explain the change in the appearance of the central maximum fringe, as well as the fringes away from this central position.

[5]

[7 marks]

Question 2c

In a third experiment, the white light is replaced by orange light of wavelength 600 nm. The double-slit has a separation of 0.350 mm and the screen is 6.35 m away.

Calculate the distance between the central and first maximum as seen on the screen.

[2 marks]

Question 2d

The light source is now changed to a blue LED of wavelength 450 nm.

Explain the features of the interference pattern that will now be observed on the screen.

[3 marks]

Question 3a

A teacher explains a diffraction investigation to a group of physics students.

'In this investigation a laser will be used to shine monochromatic light of wavelength 581.9 nm onto a diffraction grating. The light passing through the grating will therefore be coherent.'

Define the following terms used by the teacher

(i)

Monochromatic

[1]

(ii)

Coherent

[2]

[3 marks]

Question 3b

During the investigation, a second-order maximum is produced at an angle of 35° measured from the normal to the grating.

(i)

Calculate the number of lines per metre on the grating.

[3]

(ii)

Calculate the highest order which is observable.

[3]

[6 marks]

Question 3c

Calculate the angular separation between the highest order and second order maxima.

[3 marks]

Question 3d

The investigation is repeated using the same grating but a different monochromatic source.

In this experiment, the second-order maximum is observed at an angle of 25.7° .

Calculate the wavelength of this second source.

[2 marks]

Model Answers: Medium

1a

(a) Calculate the distance between the two slits:

List the known quantities:

- Wavelength, $\lambda = 650 \text{ nm} = 650 \times 10^{-9} \text{ m}$
- Angle of diffraction, $\theta = 5.00 \times 10^{-4} \text{ rad}$

Write an expression for θ using the small-angle approximation:

- From Fig. 1.1: $\tan \theta = \frac{s}{D}$
- At small angles, $\tan \theta \approx \theta$
- Therefore, $\tan \theta = \frac{s}{D} \Rightarrow \theta = \frac{s}{D}$ [1 mark]

Combine the equation for Young's Double slit with the small-angle approximation:

- Fringe spacing: $s = \frac{\lambda D}{d}$
- From the small-angle approximation: $s = \theta D$
- So, $\theta D = \frac{\lambda D}{d} \Rightarrow \theta = \frac{\lambda}{d}$ [1 mark]

Rearrange and substitute the values into the combined equation:

- $\theta = \frac{\lambda}{d} \Rightarrow d = \frac{\lambda}{\theta}$
- Distance between the two slits, $d = \frac{650 \times 10^{-9}}{5 \times 10^{-4}} = 1.3 \times 10^{-3} \text{ m}$ [1 mark]

[Total: 3 marks]

1b

(b)

(i) A bright fringe will occur at A due to:

- Constructive interference; [1 mark]

The conditions for constructive interference are:

- Light (coming from S_1 and S_2) must be coherent
AND
- The path difference (between S_1A and S_2A) must be a whole number of wavelengths; [1 mark]

(ii) State the equation for the distance S_2Y :

- $S_2Y = n\lambda$ (where n is an integer); [1 mark]

[Total: 3 marks]

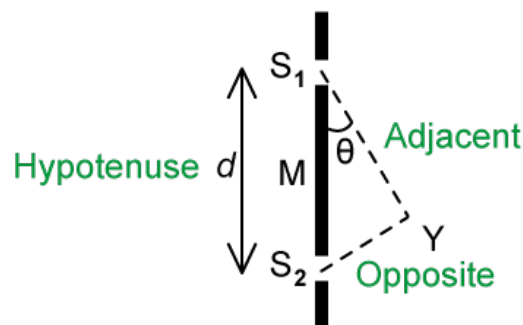
Remember constructive interference occurs when waves meet in phase. The phase difference between them is a whole number of wavelengths, so the same place on each wave will meet.

1c

(c)

(i) Deduce an expression for θ in terms of S_2Y and d :

Consider the trigonometry of the triangle S_1S_2Y :



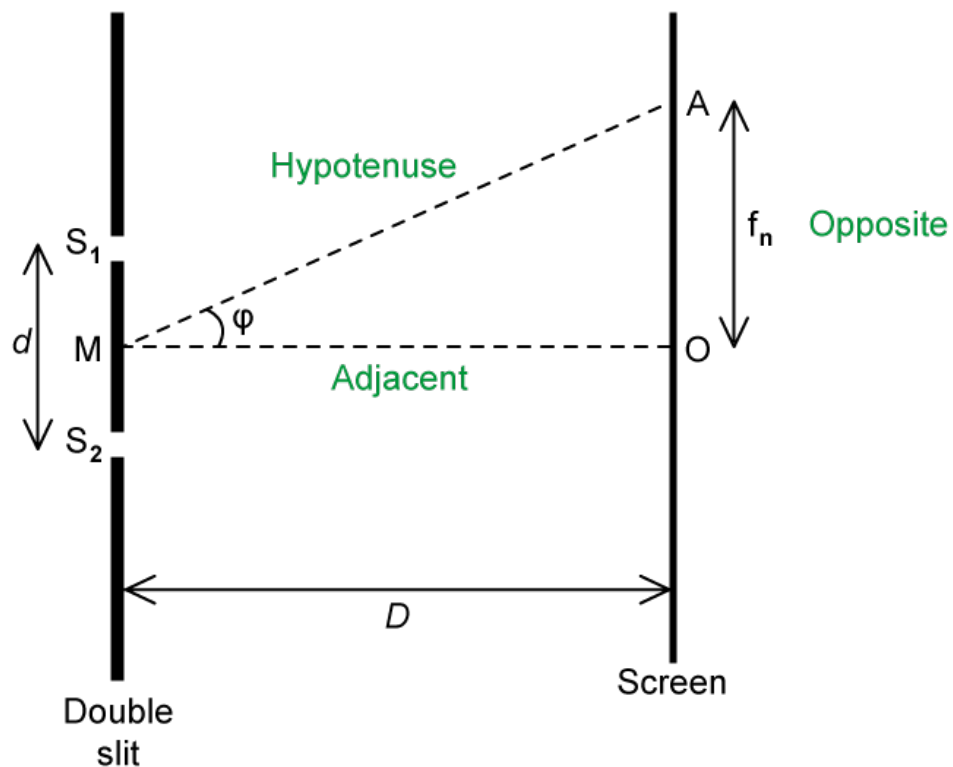
- $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{S_2 Y}{d}$ [1 mark]

Use the small-angle approximation for $\sin\theta \approx \theta$:

- $\theta = \frac{S_2 Y}{d}$ [1 mark]

(ii) Deduce an expression for ϕ in terms of D and f_n :

Consider the trigonometry of the triangle MOA:



- $\tan\phi = \frac{\text{opposite}}{\text{adjacent}} = \frac{f_n}{D}$ [1 mark]

Use small angle approximation for $\tan\phi \approx \phi$:

- $\phi = \frac{f_n}{D}$ [1 mark]

[Total: 4 marks]

1d

(d) Calculate the wavelength of the laser light:

- $S_2Y = n\lambda$ (from **(b)**)
- $\theta = \frac{S_2Y}{d}$ (from **(c)**)

Combine the equations for path difference and angle of diffraction:

- $d\theta = S_2Y = n\lambda$
- So, $\lambda = \frac{d\theta}{n}$ [1 mark]

Substitute in the known values to calculate the wavelength:

- $\lambda = \frac{d\theta}{n} = \frac{(1.3 \times 10^{-3}) \times (3.7 \times 10^{-3})}{9} = 5.34 \times 10^{-7} \text{ m}$ [1 mark]

[Total: 2 marks]

2a

(a)

(i) The appearance of the bright fringes is because:

- Bright fringes are darker; [1 mark]
- The resultant amplitude is less; [1 mark]
- Due to the reduction in the intensity of the light causing the interference; [1 mark]

(ii) The appearance of the dark fringes is because:

- Dark fringes are brighter; [1 mark]
- So, the resultant amplitude is no longer zero; [1 mark]
- Complete destructive interference from light sources of different intensities do not cancel each other out; [1 mark]

[Total: 6 marks]

To achieve full marks full descriptions of both situations with reasons must be given. There must be a reference in each reason to the resultant amplitude.

2b

(b)

(i) State the change in the appearance of the fringes and explain why:

- Fringes are closer together; [1 mark]
- Wavelength of orange light is shorter than green light **OR** frequency of orange light is higher than green light; [1 mark]

(ii) State the change in the appearance of the central fringe and explain why:

- Central fringe is white; [1 mark]
- All wavelengths interfere to give constructive interference at the central bright fringe; [1 mark]

State the change in the appearance of the outer fringes and explain why:

- Outer fringes are coloured; [1 mark]
- The inner edge of each fringe is blue and the outer edge is red; [1 mark]
- Constructive interference occurs at different positions, so very far from the central maxima no fringes are observed; [1 mark]

[Total: 7 marks]

You should have a rough idea of the wavelength of each colour, and whether the wavelength is more or less than that of another colour. Picturing the full spectrum of electromagnetic light and recalling that red comes at the long wavelength, low energy end is a useful memory hook.

2c

(c) Calculate the distance between the central and first maximum as seen on the screen:

List the known quantities:

- Central fringe is white; [1 mark]
- All wavelengths interfere to give constructive interference at the central bright fringe; [1 mark]

State the change in the appearance of the outer fringes and explain why:

- Outer fringes are coloured; [1 mark]
- The inner edge of each fringe is blue and the outer edge is red; [1 mark]
- Constructive interference occurs at different positions, so very far from the central maxima no fringes are observed; [1 mark]

[Total: 7 marks]

You should have a rough idea of the wavelength of each colour, and whether the wavelength is more or less than that of another colour. Picturing the full spectrum of electromagnetic light and recalling that red comes at the long wavelength, low energy end is a useful memory hook.

2c

(c) Calculate the distance between the central and first maximum as seen on the screen:

List the known quantities:

- Wavelength of light, $\lambda = 600 \text{ nm} = 600 \times 10^{-9} \text{ m}$
- Slit separation, $d = 0.350 \text{ mm} = 0.35 \times 10^{-3} \text{ m}$
- Distance from slit to screen, $D = 6.35 \text{ m}$

Substitute the known values into Young's Double Slit equation:

- Fringe separation, $s = \frac{\lambda D}{d}$
- $s = \frac{(600 \times 10^{-9}) \times 6.35}{(0.35 \times 10^{-3})} = 0.0109 \text{ m}$ [1 mark]

Explain your answer:

- The fringe separation is the distance between successive bright fringes
- (Hence) the distance between the central and first maximum is equal to the fringe separation; [1 mark]

[Total: 2 marks]

2d

(d) On the screen, there will be:

- No interference pattern; [1 mark]

Because:

- Light from the LED is not coherent; [1 mark]
- The phase difference between the light from the two slits will be continuously changing (hence interference cannot occur); [1 mark]

[Total: 3 marks]

Remembering the conditions for constructive and destructive interference in Young's Double Slit experiment is important.

Light must also be coherent (same frequency and constant phase difference) and monochromatic (contains light of just one wavelength).

3a

(a)

(i) Monochromatic is defined as:

- Having only one wavelength; [1 mark]

(ii) Coherent is defined as:

- Having the same frequency; [1 mark]
- And constant phase difference; [1 mark]

[Total: 3 marks]

3b

(b)

(i) Calculate the number of lines per metre on the grating:

List the known quantities:

- Wavelength, $\lambda = 581.9 \text{ nm} = 581.9 \times 10^{-9} \text{ m}$
- Second order angle, $\theta = 35^\circ$
- Order, $n = 2$

Spacing between slits, d :

- $n\lambda = d \sin\theta$
- $d = \frac{n\lambda}{\sin\theta} = \frac{2 \times (581.9 \times 10^{-9})}{\sin 35} \text{ [1 mark]}$
- $d = 2.029 \times 10^{-6} \text{ m [1 mark]}$

Calculate the number of lines per metre:

- Number of lines per metre $N = \frac{1}{d} = \frac{1}{2.029 \times 10^{-6}}$
- $N = 4.93 \times 10^5 \text{ m}^{-1} \text{ [1 mark]}$

(ii) Calculate the highest order which is observable:

- The highest order would be at $\theta = 90^\circ / \sin\theta = 1 \text{ [1 mark]}$
- $n = \frac{d}{\lambda} = \frac{2.029 \times 10^{-6}}{581.9 \times 10^{-9}} = 3.49 \text{ [1 mark]}$
- The highest order is the third / $n = 3 \text{ [1 mark]}$

[Total: 6 marks]

3c

(c) Calculate the angular separation between the second and third maxima:

List the known quantities:

- Highest order, $n = 3$
- Second order angle, $\theta_2 = 35^\circ$
- Wavelength, $\lambda = 9 \text{ nm}$

Calculate the third-order angle:

- $\sin \theta_3 = \frac{n\lambda}{d} = \frac{3 \times (581.9 \times 10^{-9})}{2.029 \times 10^{-6}} \text{ [1 mark]}$
- $\theta_3 = 59.36^\circ \text{ [1 mark]}$

Calculate the angular separation:

- $\Delta\theta = \theta_3 - \theta_2$
- $\Delta\theta = 59 - 35 = 24^\circ \text{ (2 s.f.) [1 mark]}$

[Total: 3 marks]

3d

(d) Calculate the wavelength of this second source:

List the known quantities:

- Order, $n = 2$
- Second order angle, $\theta_2 = 25.7^\circ$
- Slit separation, $d = 2.029 \times 10^{-6} \text{ m}$

Calculate the wavelength:

- $n\lambda = d \sin\theta$
- $\lambda_{\text{new}} = \frac{d \sin\theta}{n} = \frac{(2.029 \times 10^{-6}) \times \sin 25.7}{2} \text{ [1 mark]}$
- $\lambda_{\text{new}} = 4.40 \times 10^{-7} \text{ m} = 440 \text{ nm [1 mark]}$

[Total: 2 marks]